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Nuclear Fission

Energy is released whenever a nucleus emits nuclear radiation. However, the amount of energy released during a nuclear decay is very small compared to the amount of energy that can be released during nuclear fission.

Recall that nuclear fission is the process of splitting a nucleus into two or more smaller nuclei. Scientists can do this by bombarding the larger nucleus with neutrons. **Figure 8** shows a nuclear fission reaction for uranium-235. Other isotopes, including plutonium-239 and uranium-233, also undergo nuclear fission. The nuclear bomb that was dropped on Hiroshima during World War II was powered by the fission of uranium-235.

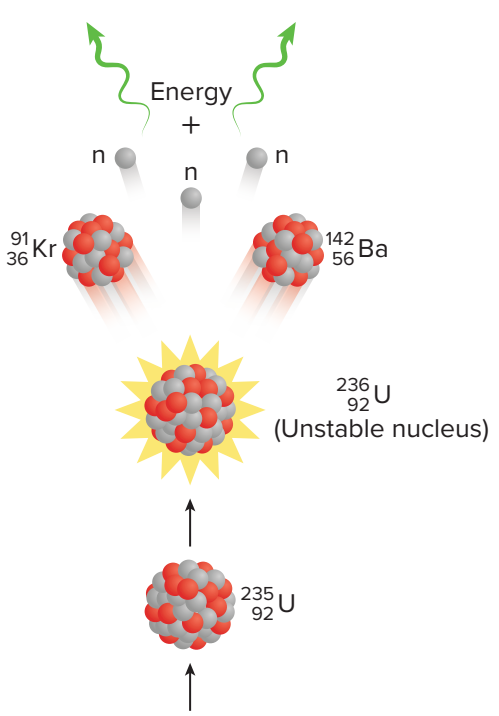


Figure 8 Some large nuclei split apart when they absorb a neutron. Here, a neutron splits a uranium-235 nucleus into two smaller nuclei and three more neutrons.

Predict *whether carbon-12 would be likely to undergo nuclear fission.*

Chain reactions

The products of a fission reaction usually include several neutrons in addition to the smaller nuclei. These neutrons can then strike other nuclei in the sample, causing them to split as well. These reactions then release more neutrons, causing additional nuclei to split. A **chain reaction** is a series of repeated fission reactions caused by the release of additional neutrons in every fission. A chain reaction is shown in **Figure 9.**

If a chain reaction is uncontrolled, an enormous amount of energy is released in an instant. This is what happens when a nuclear fission bomb is detonated. However, a chain reaction can be controlled by adding materials that absorb neutrons. If enough neutrons are absorbed, the reaction will be controlled. This is how a nuclear power plant operates.

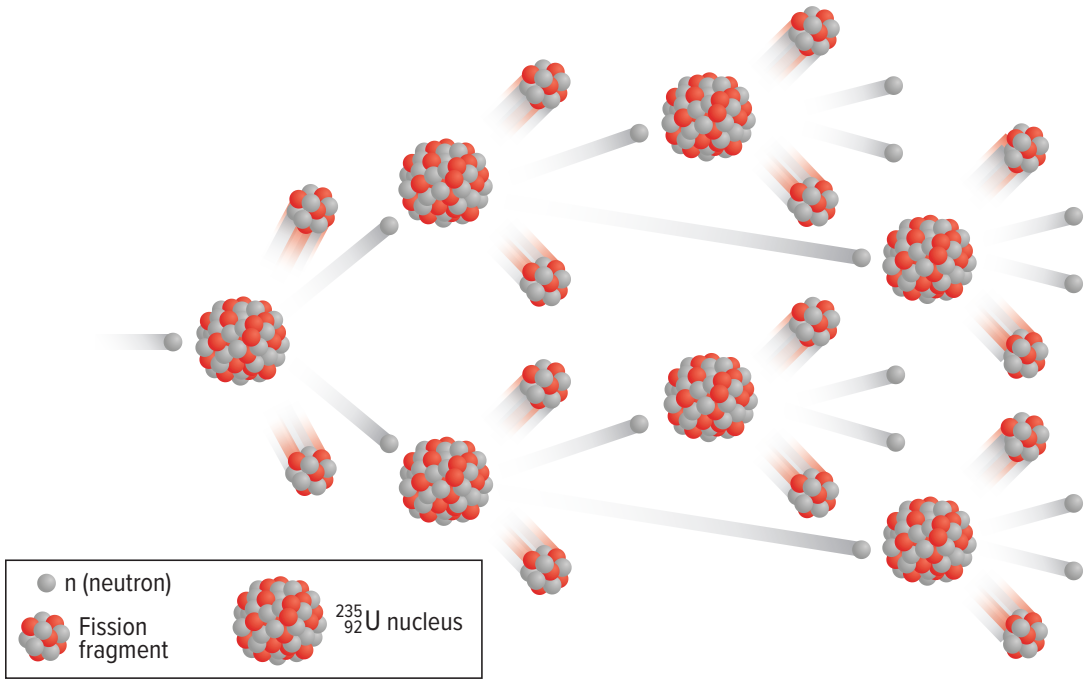


Figure 9 A chain reaction occurs when neutrons emitted from a split nucleus cause other nuclei to split and emit additional neutrons. An uncontrolled chain reaction occurs when a nuclear fission bomb is detonated. In a nuclear power plant, the chain reaction is controlled with rods of material that absorb some of the neutrons.

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